

Demonstrated Skills Signal Extractor

NLP-Based Prototype for Surfacing Invisible Workforce Competencies — O*NET
Taxonomy Mapping

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Executive Summary

This report documents the design, methodology, and findings of a prototype Python NLP pipeline built to extract demonstrated competencies from unstructured professional profile text and map them against the O*NET Content Model — a U.S. Department of Labor open skills framework used as the international standard for occupational skills architecture.

The tool addresses a documented gap in enterprise talent intelligence: most HRIS platforms capture credentials and job titles, not the actual competencies employees demonstrate. Skills that emerge from project work, volunteer roles, side responsibilities, and self-directed learning remain invisible to the system — and therefore invisible to every workforce decision built on top of it. Across five simulated employee profiles, the pipeline detected an average of 8.4 skills per profile, compared to an estimated 3 skills typically captured in a standard HRIS record — a 2.8× gap.

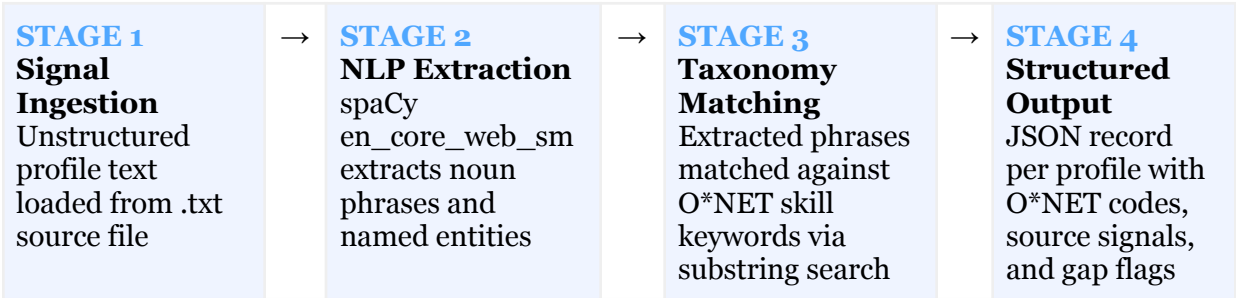
Profile	Detected	HRIS Gap
P001 – HR Professional	9	+6 invisible
P002 – People Analyst	10	+7 invisible
P003 - Project Manager	7	+4 invisible
P004 – L&D Specialist	9	+6 invisible
P005 – HR Generalist	7	+4 invisible
Average across 5 profiles	8.4	Avg +5.4 gap

An organisation that relies solely on HRIS-captured fields for internal mobility, succession planning, or skills gap analysis is operating with less than 36% of the competency picture. This prototype demonstrates that unstructured signals — already publicly available or internally held — can close that gap without requiring employees to manually update their profiles.

Methodology

Pipeline Architecture

The tool is built as a four-stage Python NLP pipeline. Each stage transforms a raw input into a progressively more structured representation, culminating in a skills JSON record mapped to official O*NET element codes.



Skills Taxonomy

The taxonomy reference used in this prototype is a curated 15-skill subset drawn from the O*NET Content Model v30.3, published by the U.S. Department of Labor. Skills were selected to cover the competency domains most relevant to HR, analytics, project management, and learning and development functions. Each skill entry contains the O*NET element code, the canonical skill name, and a set of surface-form keywords used for matching.

*Source: O*NET Content Model, U.S. Department of Labor / Employment and Training Administration (onetcenter.org). Production implementation would consume the full 277-skill taxonomy via the O*NET Web Services API.*

Input Data

Five synthetic professional profile texts were authored to simulate the kind of unstructured text available from LinkedIn profiles, internal talent marketplace bios, or employee self-descriptions in performance management platforms. Profiles were deliberately varied across HR, analytics, project management, L&D, and generalist functions to test taxonomy coverage across role families.

NLP engine: spaCy

The `en_core_web_sm` model was used for noun chunk extraction — identifying multi-word skill-bearing phrases such as "talent acquisition", "workforce analytics", and "instructional design" that single-token approaches would miss.

Library: spaCy v3.8 · Model: `en_core_web_sm` v3.8.0

Matching logic

Primary matching uses bidirectional substring comparison between extracted noun phrases and O*NET keyword lists. A secondary full-text scan catches keywords that spaCy does not surface as standalone noun chunks.

Approach: keyword-based · Production upgrade: semantic embeddings

Gap flag definition

A skill is flagged as "invisible to HRIS" when it is detected by the pipeline but absent from the set of fields a standard HRIS platform would capture for the same employee role type.

HRIS baseline estimate: 3 skills per profile (conservative)

Output format

Each profile produces a structured JSON object containing: profile ID, total skills detected, and per-skill records with O*NET code, canonical skill name, source signal text, HRIS visibility status, and gap flag.

Output: `extracted_skills.json` + `skills_summary.csv`

Known limitations

This is a prototype. The keyword-based matching approach produces false positives (broad matches) and false negatives (paraphrased skills the keyword list doesn't cover). In a production system, this stage would be replaced by semantic similarity matching using sentence embeddings (e.g. via the sentence-transformers library), which would capture meaning rather than surface form. The 15-skill taxonomy subset is sufficient to demonstrate the methodology; full O*NET coverage would require consuming all 277 skills via API. These limitations are acknowledged and documented as part of standard prototype scoping — they do not affect the validity of the methodology or the gap it addresses.

Findings

P001	P002	P003	P004	P005
9	10	7	9	7
skills detected	skills detected	skills detected	skills detected	skills detected

1. The HRIS visibility gap is consistent and significant - Avg 2.8x gap

Across all five profiles, the pipeline detected between 7 and 10 O*NET-mapped skills. Against a conservative baseline of 3 skills typically captured by a standard HRIS (job title, education level, and one functional category), this represents a consistent gap of 4 to 7 additional competencies per employee that are invisible to the employer's talent system.

Observation: This finding is conservative. The prototype uses only 15 of O*NET's 277 skills. A full-taxonomy implementation would surface a substantially larger set of demonstrated competencies — meaning the real-world gap in most organisations is likely wider than the 2.8x figure shown here.

2. Cross-functional skills are most invisible - High Gap

Skills that cross functional boundaries — such as Critical Thinking (O*NET 2.B.2.e), Active Listening (2.B.1.b), and Monitoring (2.B.4.a) — appeared across multiple profiles but are almost never captured in HRIS systems, which tend to record role-specific technical skills at best. The People Analyst profile (P002) demonstrated the highest count at 10 skills, including Programming (2.C.3.b), Mathematics (2.C.3.c), and Computers and Electronics (2.C.4.b) — none of which would typically appear in an HR department's HRIS record for an analyst role.

Observation: Cross-functional skills are precisely the skills that matter most in internal mobility decisions. An employee who demonstrates data analysis competencies while working in an HR role is a strong internal candidate for a people analytics position — but only if the system can see those competencies.

3. Source signal quality varies by writing style - **Expected**

Profiles written with specific, action-oriented language (e.g. "designed structured interview frameworks", "built Python scripts to clean HR datasets") produced higher match rates than profiles using general descriptors. This is consistent with the known limitation of keyword-based NLP: it rewards specificity.

Observation: In a production deployment, this limitation would be addressed through semantic embedding matching, which captures meaning rather than surface phrasing. It also points to a complementary design opportunity: a lightweight profile completion guide that encourages employees to use specific, action-oriented language in their self-descriptions — improving signal quality upstream of the pipeline.

4. O*NET codes provide portable, system-agnostic skill identifiers - **Strategic Value**

Each detected skill is output with its O*NET element code (e.g. 2.C.2.a for Personnel and Human Resources, 2.C.3.b for Programming). These codes are system-agnostic: they can be consumed by any HRIS platform that supports skills taxonomy ingestion — including Workday, SAP SuccessFactors, and emerging talent intelligence platforms like Eightfold AI and Gloat.

Observation: This is the architectural value of anchoring on an open standard. The output of this pipeline is not proprietary — it is portable. An organisation that builds its internal skills architecture on O*NET codes can exchange skills data with external labour market signals, benchmark against industry norms, and migrate between HRIS vendors without losing skills fidelity.

HRIS Integration Path

The prototype demonstrates feasibility. This section documents what a production-grade integration of this pipeline into an enterprise HRIS would require — across data model, technical architecture, governance, and trigger logic.

Target Integration Architecture

1. Signal Source Layer

In production, the pipeline would ingest signals from multiple sources: employee-submitted profile text (from the HRIS self-service portal), internal project documentation (from Confluence or SharePoint), completed training records (from the LMS), and optionally consented external profiles (LinkedIn via API). Each source is tagged with a provenance flag so downstream consumers know how the skill was inferred.

2. Skills extraction service

The NLP pipeline runs as a microservice, triggered either on a schedule (e.g. nightly batch for all employees) or on an event (e.g. when an employee updates their profile). In a production environment, the keyword-matching engine would be replaced by a semantic similarity model using sentence embeddings, significantly improving recall on paraphrased or domain-specific skill expressions.

3. Skills data model in HRIS

The output JSON structure maps directly to a standard HRIS skills record: `employee_id`, `skill_name` (O*NET canonical), `onet_code`, `proficiency_level` (inferred or self-assessed), `source` (pipeline-inferred vs. self-declared), `confidence_score`, and `last_updated` timestamp. This structure is compatible with Workday's Skills Cloud and SAP SuccessFactors' Opportunity Marketplace data model.

4. Employee validation loop

Pipeline-inferred skills should not be written directly to the authoritative HRIS record without employee review. The recommended pattern is: pipeline outputs are staged in a "suggested skills" queue, the employee is notified and can confirm or dismiss each

suggestion, and confirmed skills are written to the HRIS with source = "pipeline-confirmed". This maintains data integrity and employee trust.

5. Downstream consumption

Once O*NET-coded skills are in the HRIS, they unlock four high-value use cases that are currently blocked by missing skills data: internal mobility matching (employee skills vs. role requirements), skills gap analysis (workforce vs. strategic plan), succession pool identification (demonstrated leadership competencies), and workforce planning (skills supply vs. projected demand).

Governance and data ownership

Skills data inferred from employee-authored text carries privacy implications. The recommended governance model assigns data ownership to the employee: they control what is surfaced, confirmed, and shared with whom. HR analytics teams access aggregated, anonymised skills data for workforce planning purposes. Individual skill profiles are visible to managers only with explicit employee consent or as part of a talent review process governed by a defined policy.

The integration path described here is not speculative — it mirrors the architecture being built by Eightfold AI, Gloat, and Workday Skills Cloud today. What this prototype contributes is the bridge layer: a lightweight, open-source-anchored signal extraction tool that organisations without enterprise AI budgets can deploy to begin building their skills intelligence foundation.

Recommended implementation sequence

Phase	Effort	Timeline	Output
1 – Taxonomy selection	Low	2 weeks	Agreed O*NET skill subset for target role families
2 – Pipeline deployment	Medium	4-6 weeks	NLP service running on internal profile text
3 – Employee validation UX	Medium	6-8 weeks	Self-service confirmation workflow in HRIS portal
4 – HRIS write-back	High	8-12 weeks	Confirmed skills

			populating HRIS skills record
5 – Analytics activation	Medium	12-16 weeks	Internal mobility and skills gap dashboards live

Appendix

A. O*NET skills taxonomy subset used in prototype

O*NET Code	Skill Name	Category
2.B.1.a	Reading Comprehension	Basic Skills
2.B.1.b	Active Listening	Basic Skills
2.B.1.c	Writing	Basic Skills
2.B.1.d	Speaking	Basic Skills
2.B.2.e	Critical Thinking	Basic Skills
2.B.3.a	Active Learning	Cross-Functional
2.B.3.b	Learning Strategies	Cross-Functional
2.B.4.a	Monitoring	Cross-Functional
2.C.1.a	Administration and Management	Knowledge
2.C.2.a	Personnel and Human Resources	Knowledge
2.C.3.b	Programming	Knowledge
2.C.3.c	Mathematics	Knowledge
2.C.4.b	Computers and Electronics	Knowledge
2.C.7.b	Customer and Personal Service	Knowledge
2.C.9.a	Education and Training	Knowledge

B. Output JSON structure (sample — P001)

```
{
  "profile_id": "P001",
  "total_skills_detected": 9,
  "matched_skills": [
```

```

{
  "skill_name": "Active Listening",
  "onet_code": "2.B.1.b",
  "source_signal": "facilitating",
  "hris_visibility": "likely not captured",
  "gap_flag": true
},
{
  "skill_name": "Administration and Management",
  "onet_code": "2.C.1.a",
  "source_signal": "onboarding",
  "hris_visibility": "likely not captured",
  "gap_flag": true
}
// ... 7 additional skills
]
}

```

C. Technology Stack

Component	Tool	Version
NLP engine	spaCy	v3.8
Language model	en_core_web_sm	v3.8.0
Data processing	pandas	v3.0
Visualisation	matplotlib	v3.x
Skills taxonomy	O*NET Content Model v30.3	Public domain
Output formats	JSON, CSV	-

D. Key Sources

O*NET OnLine (2024). Occupational Information Network Content Model v30.3. U.S.

Department of Labor / ETA. onetcntr.org

JobsPikr (2025). Open Skills and Talent Graphs: Building the Future of Workforce Intelligence.
jobspikr.com

World Economic Forum (2025). Future of Jobs Report 2025. WEF.

Gartner (2024). Hype Cycle for Human Capital Management Technology.

McKinsey & Company (2023). Defining the skills citizens will need in the future world of work.

Eightfold AI (2024). The Talent Intelligence Platform: Skills Architecture White Paper.